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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 28th July Evening Shift

JEE Main Question Paper with Answer Key

28 Jul 2022	90	24	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	20 questions
Mathematics	19 questions
Physics	27 questions
Out of syllabus	24 questions
Answer key	at the end

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Chemistry 20 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements: One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion $\mathbf{A}$: Zero orbital overlap is an out of phase overlap.

Reason $\mathbf{R}$: It results due to different orientation / direction of approach of orbitals.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements: One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The reduction of a metal oxide is easier if the metal formed is in liquid state than solid state.

Reason $\mathbf{R}$: The value of ΔG^{Θ} becomes more on negative side as entropy is higher in liquid state than solid state.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct and R is the correct explanation of A

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Q3. MCQ (More than One Correct Answer)

Open solution

Which of the following pair is not isoelectronic species?

(At. no. Sm, 62; Er, 68; Yb, 70; Lu, 71; Eu, 63; Tb, 65; Tm, 69)

(A) Sm^{2+} and Er^{3+}

(B) Yb^{2+} and Lu^{3+}

(C) Eu^{2+} and Tb^{4+}

(D) Tb^{2+} and Tm^{4+}

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Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Permanganate titrations are not performed in presence of hydrochloric acid.

Reason R : Chlorine is formed as a consequence of oxidation of hydrochloric acid.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q5. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List - I (Complex)		List - II (Hybridization)
(A)	$Ni(CO)_4$	(I)	sp^3
(B)	$[Ni(CN)_4]^{2-}$	(II)	sp^3d^2
(C)	$[Co(CN)_6]^{3-}$	(III)	d^2sp^3
(D)	$[CoF_6]^{3-}$	(IV)	dsp^2

Choose the correct answer from the options given below :

(A) A-IV, B-I, C-III, D-II

(B) A-I, B-IV, C-III, D-II

(C) A-I, B-IV, C-II, D-III

(D) A-IV, B-I, C-II, D-III

Q6. MCQ (Single Correct Answer)

Open solution

Dinitrogen and dioxygen, the main constituents of air do not react with each other in atmosphere to form oxides of nitrogen because :

(A) N_2 is unreactive in the condition of atmosphere.

(B) Oxides of nitrogen are unstable.

(C) Reaction between them can occur in the presence of a catalyst.

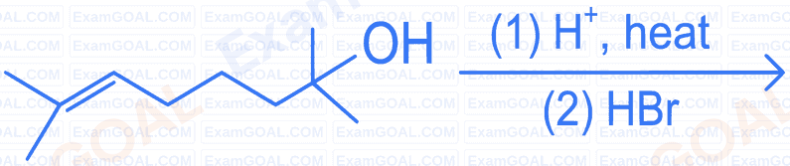
(D) The reaction is endothermic and require very high temperature.

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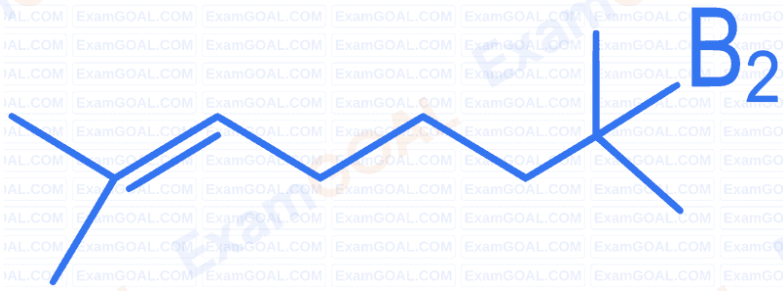
Q7. MCQ (Single Correct Answer)

Open solution

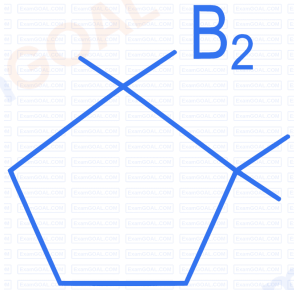
The major product in the given reaction is



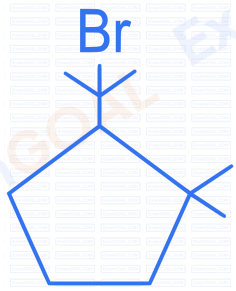
(A)



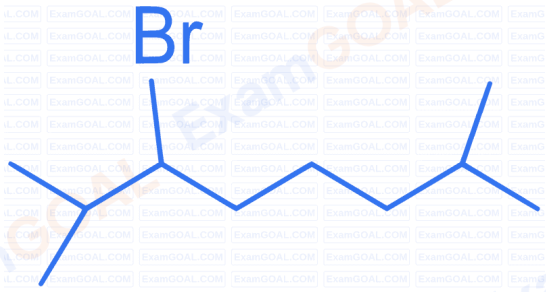
(B)



(C)



(D)



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Q8. MCQ (Single Correct Answer)

Open solution

Arrange the following in increasing order of reactivity towards nitration

A. p-xylene

B. bromobenzene

C. mesitylene

D. nitrobenzene

E. benzene

Choose the correct answer from the options given below

(A) C < D < E < A < B

(B) D < B < E < A < C

(C) D < C < E < A < B

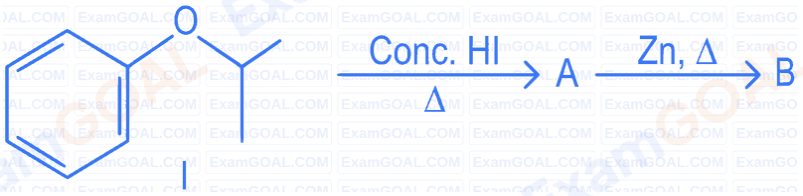
(D) C < D < E < B < A

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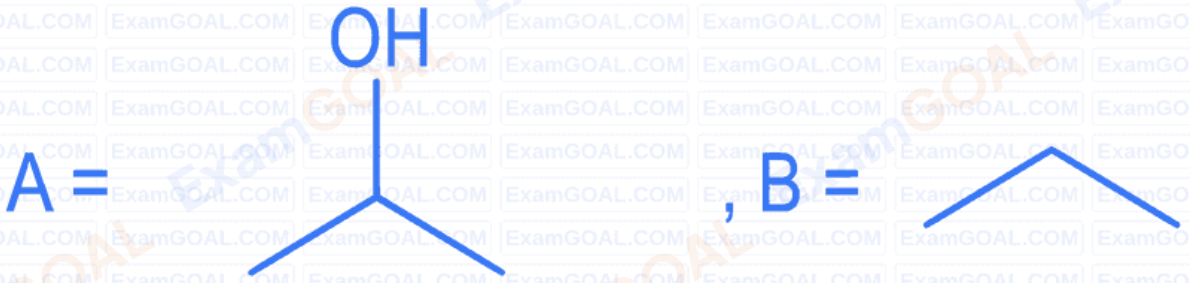
Q9. MCQ (Single Correct Answer)

Open solution

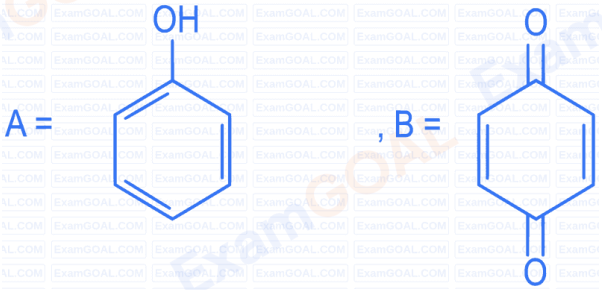
Compound I is heated with Conc. HI to give a hydroxy compound A which is further heated with Zn dust to give compound B. Identify A and B.



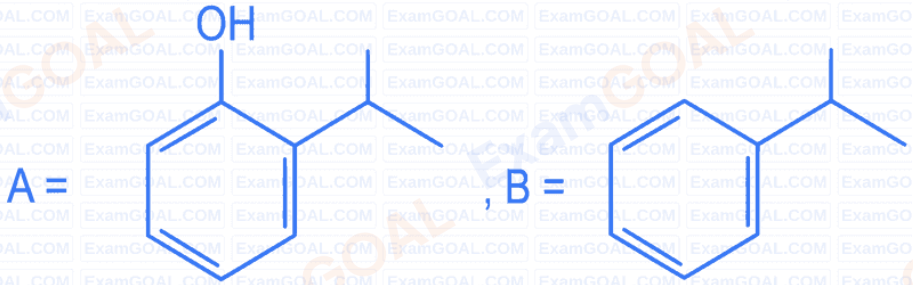
(A)



(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : Aniline on nitration yields ortho, meta & para nitro derivatives of aniline.

Reason R : Nitrating mixture is a strong acidic mixture.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Thin layer chromatography is an adsorption chromatography.

Reason R : A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

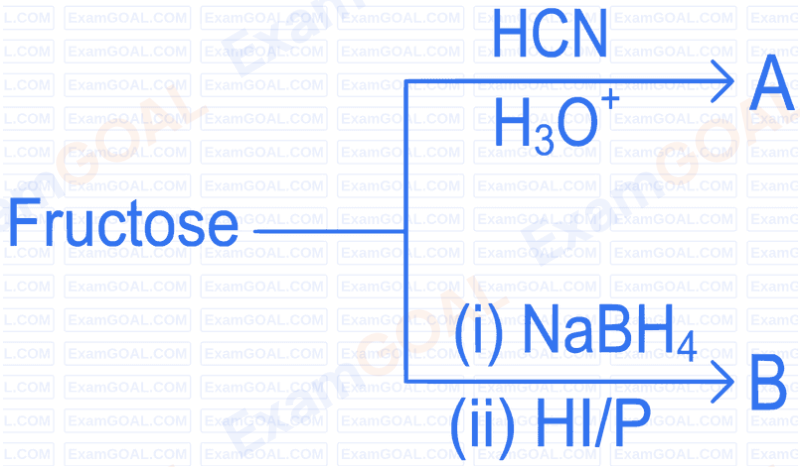
(D) A is false but R is true

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Q12. MCQ (Single Correct Answer)

Open solution

The formulas of A and B for the following reaction sequence



are

(A) $A = C_7H_{14}O_8, \quad B = C_6H_{14}$

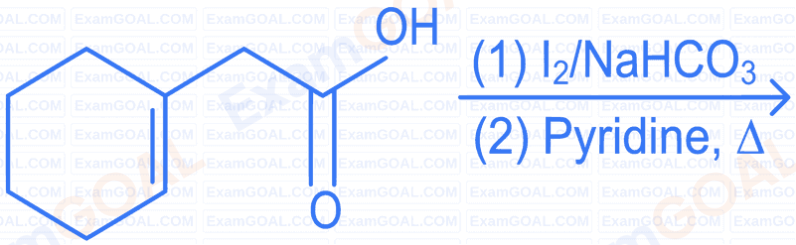
(B) $A = C_7H_{13}O_7, \quad B = C_7H_{14}O$

(C) $A = C_7H_{12}O_8, \quad B = C_6H_{14}$

(D) $A = C_7H_{14}O_8, \quad B = C_6H_{14}O_6$

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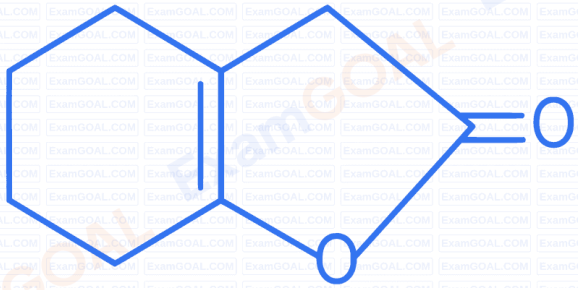
Q13. MCQ (Single Correct Answer)



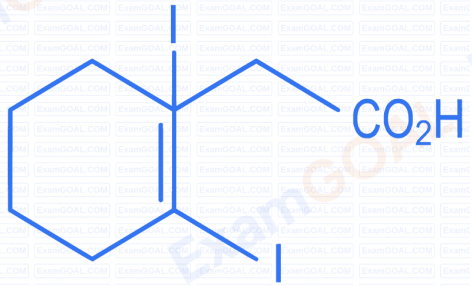
Open solution

Find out the major product for the above reaction.

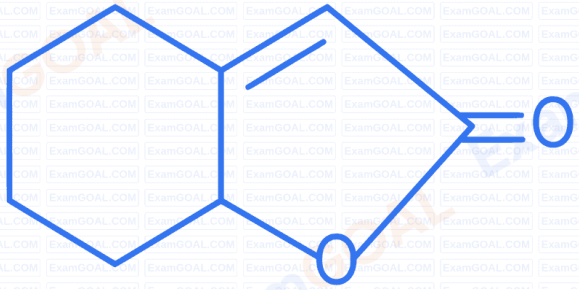
(A)



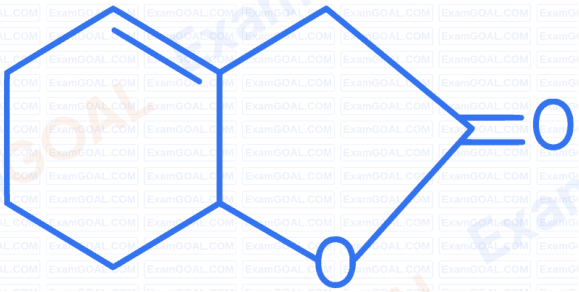
(B)



(C)



(D)



Q14. Numerical

2L of 0.2M H_2SO_4 is reacted with 2L of 0.1M NaOH solution, the molarity of the resulting product Na_2SO_4 in the solution is _____ millimolar. (Nearest integer)

Open solution

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Q15. Numerical

If the wavelength for an electron emitted from H-atom is $3.3 \times 10^{-10}\ \mathrm{m}$, then energy absorbed by the electron in its ground state compared to minimum energy required for its escape from the atom, is _____ times. (Nearest integer)

[Given : $\mathrm{h} = 6.626 \times 10^{-34}\ \mathrm{J\,s}$]

Mass of electron $= 9.1 \times 10^{-31}\ \mathrm{kg}$

Open solution

Q16. Numerical

A gaseous mixture of two substances A and B, under a total pressure of 0.8 atm is in equilibrium with an ideal liquid solution. The mole fraction of substance A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is _____ atm. (Nearest integer)

Open solution

Q17. Numerical

At $600\ \mathrm{K}, 2\ \mathrm{mol}$ of NO are mixed with $1\ \mathrm{mol}$ of $\mathrm{O_2}$.

$2\mathrm{NO}_{(g)} + \mathrm{O_2(g)} \rightleftarrows 2\mathrm{NO_2(g)}$

The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that $0.6\ \mathrm{mol}$ of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer)

Open solution

Q18. Numerical

A sample of $0.125\ \mathrm{g}$ of an organic compound when analyzed by Duma's method yields $22.78\ \mathrm{mL}$ of nitrogen gas collected over KOH solution at $280\ \mathrm{K}$ and $759\ \mathrm{mm\ Hg}$. The percentage of nitrogen in the given organic compound is _____. (Nearest integer)

Given :

(a) The vapour pressure of water of $280\ \mathrm{K}$ is $14.2\ \mathrm{mm\ Hg}$.

(b) $\mathrm{R} = 0.082\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$

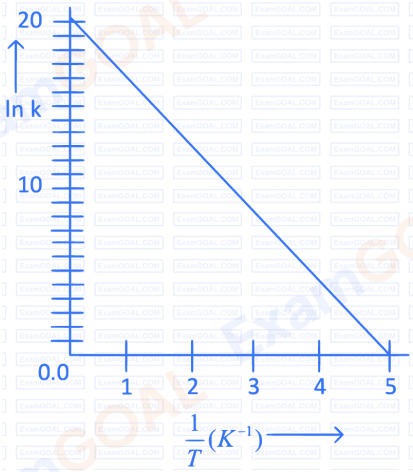
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Q19. Numerical

For a reaction, given below is the graph of $\ln k$ vs $\frac{1}{T}$. The activation energy for the reaction is equal to _____ cal mol^{-1}. (nearest integer)

(Given : $\text{R} = 2\,\text{cal K}^{-1}\ \text{mol}^{-1}$)



Open solution

Q20. Numerical

Among the following the number of state variables is _____.

Internal energy (U)

Volume (V)

Heat (q)

Enthalpy (H)

Open solution

Mathematics 19 questions

Q21. MCQ (Single Correct Answer)

Let $S = \left\{x \in [-6,3] - \{-2,2\} : \frac{|x+3|-1}{|x|-2} \geq 0\right\}$ and $T = \{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\}$.

Then the number of elements in $\text{S} \cap \text{T}$ is :

(A) 7 (B) 5 (C) 4 (D) 3

Open solution

Q22. MCQ (Single Correct Answer)

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a+b-2)x + (a+b+2) = 0$ are :

(A) non-real complex numbers (B) real and both negative

(C) real and both positive (D) real and exactly one of them is positive

Open solution

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Q23. MCQ (Single Correct Answer)

Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?

(A) $\mathrm{A}^4 - \mathrm{B}^4$ is a symmetric matrix

(B) $\mathrm{AB} - \mathrm{BA}$ is a symmetric matrix

(C) $\mathrm{B}^5 - \mathrm{A}^5$ is a skew-symmetric matrix

(D) $\mathrm{AB} + \mathrm{BA}$ is a skew-symmetric matrix

Open solution

Q24. MCQ (Single Correct Answer)

Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3, f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to :

(A) -4 (B) $\frac{13}{2}$ (C) $\frac{23}{2}$ (D) 4

Open solution

Q25. MCQ (Single Correct Answer)

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$f(x) = \lim\limits_{n\to\infty} \frac{\cos(2\pi x) - x^{2n}\sin(x-1)}{1 + x^{2n+1} - x^{2n}}$ is continuous for all x in :

(A) $R - \{-1\}$ (B) $\mathbb{R} - \{-1, 1\}$ (C) $R - \{1\}$ (D) $R - \{0\}$

Open solution

Q26. MCQ (Single Correct Answer)

The function $f(x) = xe^{x(1-x)}, x \in \mathbb{R}$, is :

(A) increasing in $\left(-\frac{1}{2}, 1\right)$

(B) decreasing in $\left(\frac{1}{2}, 2\right)$

(C) increasing in $\left(-1, -\frac{1}{2}\right)$

(D) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Open solution

Q27. MCQ (Single Correct Answer)

The sum of the absolute maximum and absolute minimum values of the function $f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0, \pi]$ is :

(A) 0

(B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$

(C) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$

(D) $\frac{-\pi}{12}$

Open solution

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Q28. MCQ (Single Correct Answer)

Let $x(t) = 2\sqrt{2}\cos t\sqrt{\sin 2t}$ and

$y(t) = 2\sqrt{2}\sin t\sqrt{\sin 2t}, t \in \left(0, \frac{\pi}{2}\right)$.

Then $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ at $t = \frac{\pi}{4}$ is equal to :

(A) $\frac{-2\sqrt{2}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{-2}{3}$

Open solution

Q29. MCQ (Single Correct Answer)

Let $I_n(x) = \int_0^x \frac{1}{(t^2+5)^n}dt, n = 1, 2, 3, \ldots$. Then :

(A) $50I_6 - 9I_5 = xI_5'$ (B) $50I_6 - 11I_5 = xI_5'$

(C) $50I_6 - 9I_5 = I_5'$ (D) $50I_6 - 11I_5 = I_5'$

Open solution

Q30. MCQ (Single Correct Answer)

The area enclosed by the curves $y = \log_e\left(x + e^2\right), x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is:

(A) $2 + e - \log_e 2$ (B) $1 + e - \log_e 2$

(C) $e - \log_e 2$ (D) $1 + \log_e 2$

Open solution

Q31. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{1/2}, x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}\,y(8)$ is equal to :

(A) $11 + 6\log_e 3$ (B) 19 (C) $12 - 2\log_e 3$ (D) $19 - 6\log_e 3$

Open solution

Q32. MCQ (Single Correct Answer)

Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through the point $(2\sqrt{2}, -2\sqrt{2})$. A parabola is drawn whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H. If the length of the latus rectum of the parabola is e times the length of the latus rectum of H, where e is the eccentricity of H, then which of the following points lies on the parabola?

(A) $(2\sqrt{3}, 3\sqrt{2})$ (B) $(3\sqrt{3}, -6\sqrt{2})$ (C) $(\sqrt{3}, -\sqrt{6})$ (D) $(3\sqrt{6}, 6\sqrt{2})$

Open solution

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Q33. MCQ (Single Correct Answer)

Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, $(b > 1)$ is acute. Then S is equal to :

(A) $\left(-\infty, -\frac{4}{3}\right)$

(B) Φ

(C) $\left(-\frac{4}{3}, 0\right)$

(D) $\left(\frac{12}{7}, \infty\right)$

Open solution

Q34. MCQ (Single Correct Answer)

Let A and B be two events such that $P(B \mid A) = \frac{2}{5}, P(A \mid B) = \frac{1}{7}$ and $P(A \cap B) = \frac{1}{9}$. Consider

(S1) $P(A' \cup B) = \frac{5}{6}$,

(S2) $P(A' \cap B') = \frac{1}{18}$

Then :

(A) Both (S1) and (S2) are true

(B) Both (S1) and (S2) are false

(C) Only (S1) is true

(D) Only (S2) is true

Open solution

Q35. Numerical

Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}} + \beta x\right)^4, (1 - 3\beta x)^2$ and $\left(1 - \frac{\beta}{2}x\right)^6, \beta > 0$, respectively form the first three terms of an A.P. If d is the common difference of this A.P. , then $50 - \frac{2d}{\beta^2}$ is equal to _____.

Open solution

Q36. Numerical

A class contains b boys and g girls. If the number of ways of selecting 3 boys and 2 girls from the class is 168 , then $\mathrm{b} + 3\,\mathrm{g}$ is equal to _____.

Open solution

Q37. Numerical

Let $\mathrm{z} = a + ib, b \neq 0$ be complex numbers satisfying $z^2 = \bar{z} \cdot 2^{1-|z|}$. Then the least value of $n \in N$, such that $z^n = (z+1)^n$, is equal to _____.

Open solution

Q38. Numerical

A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X, then $100\sigma^2$ is equal to _____.

Open solution

Q39. Numerical

The value of the integral $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ is equal to _____.

Open solution

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Physics 27 questions

Q40. MCQ (Single Correct Answer)

Consider the efficiency of Carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta}\log_e \frac{\beta x}{kT}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option :

Open solution

(A) Dimensions of β is same as that of force.

(B) Dimensions of $\alpha^{-1}x$ is same as that of energy.

(C) Dimensions of $\eta^{-1}\sin\theta$ is same as that of $\alpha\beta$.

(D) Dimensions of α is same as that of β.

Q41. MCQ (Single Correct Answer)

At time $t = 0$ a particle starts travelling from a height $7\hat{z}\,\text{cm}$ in a plane keeping z coordinate constant. At any instant of time its position along the $\hat{x}$ and $\hat{y}$ directions are defined as 3t and 5t^3 respectively. At t = 1s acceleration of the particle will be

Open solution

(A) $-30\hat{y}$ (B) $30\hat{y}$ (C) $3\hat{x} + 15\hat{y}$ (D) $3\hat{x} + 15\hat{y} + 7\hat{z}$

Q42. MCQ (Single Correct Answer)

A pressure-pump has a horizontal tube of cross sectional area $10\,\text{cm}^2$ for the outflow of water at a speed of $20\,\text{m/s}$. The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is :

Open solution

[given: density of water $= 1000\,\text{kg/m}^3$]

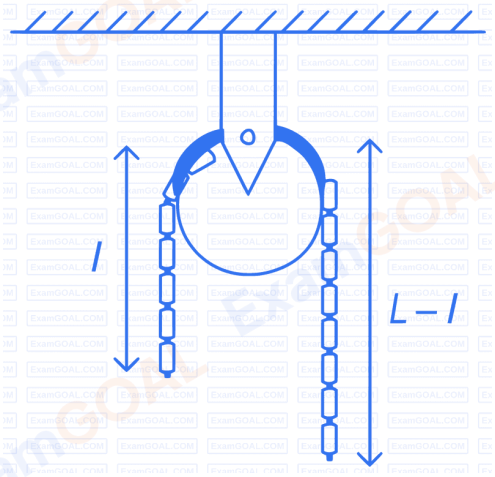
(A) 300 N (B) 500 N (C) 250 N (D) 400 N

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Q43. MCQ (Single Correct Answer)

Open solution

A uniform metal chain of mass m and length 'L' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'l' hanging on one side and rest of its length '$\mathrm{L}-l$' is hanging on the other side of the pulley. At a certain point of time, when $l=\frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is _____.

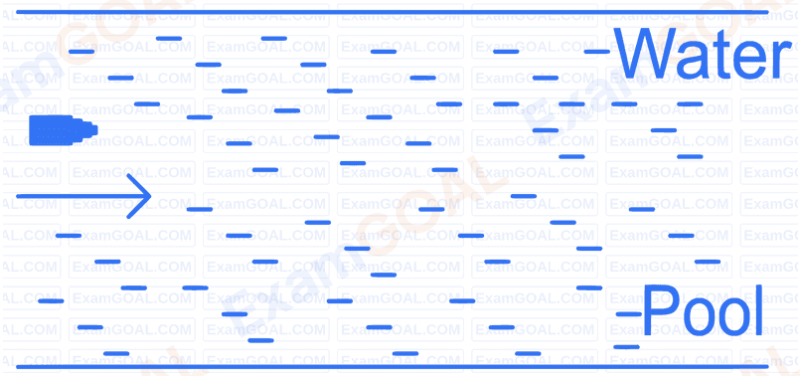


(A) 6 (B) 2 (C) 1.5 (D) 4

Q44. MCQ (Single Correct Answer)

Open solution

A bullet of mass $200\ \mathrm{g}$ having initial kinetic energy $90\ \mathrm{J}$ is shot inside a long swimming pool as shown in the figure. If its kinetic energy reduces to $40\ \mathrm{J}$ within $1\ \mathrm{s}$, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



(A) 45 m (B) 90 m (C) 125 m (D) 25 m

Q45. MCQ (Single Correct Answer)

Open solution

Assume there are two identical simple pendulum clocks. Clock - 1 is placed on the earth and Clock - 2 is placed on a space station located at a height h above the earth surface. Clock - 1 and Clock - 2 operate at time periods 4 s and 6 s respectively. Then the value of h is -

(consider radius of earth $R_E=6400\ \mathrm{km}$ and g on earth $10\ \mathrm{m/s^2}$)

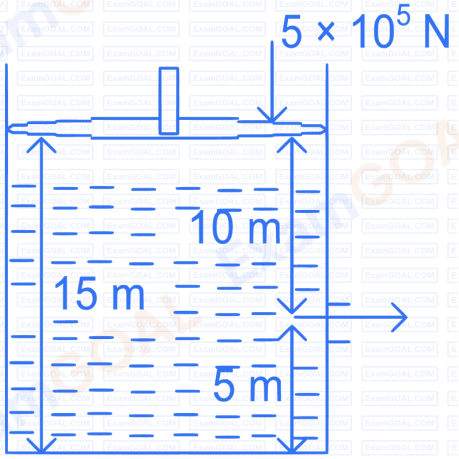
(A) 1200 km (B) 1600 km (C) 3200 km (D) 4800 km

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Q46. MCQ (Single Correct Answer)

Open solution

Consider a cylindrical tank of radius $1\ \mathrm{m}$ is filled with water. The top surface of water is at $15\ \mathrm{m}$ from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of $5\ \mathrm{m}$ from the bottom. A force of $5 \times 10^5\ \mathrm{N}$ is applied on the top surface of water using a piston. The speed of efflux from the hole will be : (given atmospheric pressure $\mathrm{P_A} = 1.01 \times 10^5\ \mathrm{Pa}$, density of water $\rho_{\mathrm{W}} = 1000\ \mathrm{kg/m^3}$ and gravitational acceleration $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 11.6 m/s (B) 10.8 m/s (C) 17.8 m/s (D) 14.4 m/s

Q47. MCQ (Single Correct Answer)

Open solution

A vessel contains $14\ \mathrm{g}$ of nitrogen gas at a temperature of $27^{\circ}\mathrm{C}$. The amount of heat to be transferred to the gas to double the r.m.s speed of its molecules will be :

Take $\mathrm{R} = 8.32\ \mathrm{J\ mol^{-1}\ k^{-1}}$.

(A) 2229 J (B) 5616 J (C) 9360 J (D) 13,104 J

Q48. MCQ (Single Correct Answer)

Open solution

A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}\ \mathrm{d}$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be :

(Given $\mathrm{C_0}$ = capacitance of capacitor with air as medium between plates.)

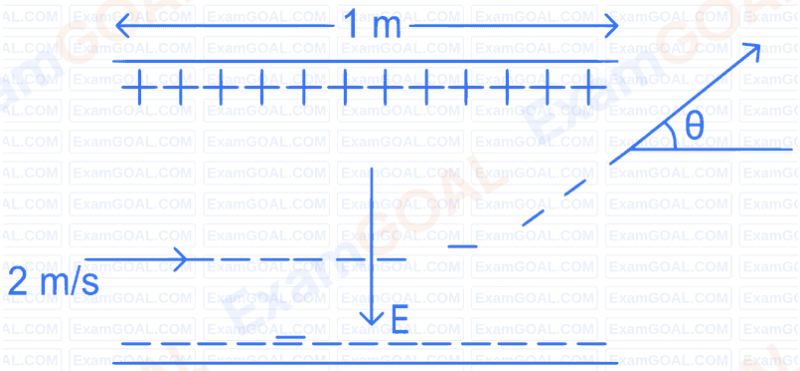
(A) $\frac{4KC_0}{3+K}$ (B) $\frac{3KC_0}{3+K}$ (C) $\frac{3+K}{4KC_0}$ (D) $\frac{K}{4+K}$

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Q49. MCQ (Single Correct Answer)

Open solution

A uniform electric field $\mathrm{E} = (8\,\mathrm{m/e})\,\mathrm{V/m}$ is created between two parallel plates of length $1\,\mathrm{m}$ as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of $2\,\mathrm{m/s}$. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be _____.



(A) $\tan^{-1}(4)$

(B) $\tan^{-1}(2)$

(C) $\tan^{-1}\left(\frac{1}{3}\right)$

(D) $\tan^{-1}(3)$

Q50. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : A uniform wire of resistance $80\,\Omega$ is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be $5\,\Omega$.

Statement II: Two resistances 2R and 3R are connected in parallel in an electric circuit. The value of thermal energy developed in 3R and 2R will be in the ratio $3 : 2$.

In the light of the above statements, choose the most appropriate answer from the option given below

(A) Both statement I and statement II are correct

(B) Both statement I and statement II are incorrect

(C) Statement I is correct but statement II is incorrect

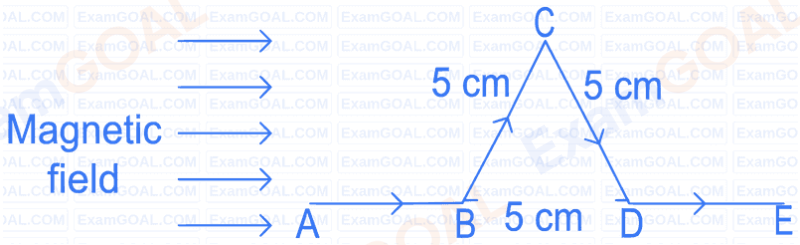
(D) Statement I is incorrect but statement II is correct

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Q51. MCQ (Single Correct Answer)

A triangular shaped wire carrying $10\ \mathrm{A}$ current is placed in a uniform magnetic field of $0.5\ \mathrm{T}$, as shown in figure. The magnetic force on segment CD is

(Given $\mathrm{BC} = \mathrm{CD} = \mathrm{BD} = 5\ \mathrm{cm}$.)



Open solution

(A) 0.126 N (B) 0.312 N (C) 0.216 N (D) 0.245 N

Q52. MCQ (Single Correct Answer)

The magnetic field at the center of current carrying circular loop is B_1. The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2. The value of B_1/B_2 will be

Open solution

(A) 9 : 4 (B) $12 : \sqrt{5}$ (C) 8 : 1 (D) $5 : \sqrt{3}$

Q53. MCQ (Single Correct Answer)

A transformer operating at primary voltage $8\ \mathrm{kV}$ and secondary voltage $160\ \mathrm{V}$ serves a load of $80\ \mathrm{kW}$. Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be

Open solution

(A) $800\ \Omega$ and $1.06\ \Omega$ (B) $10\ \Omega$ and $500\ \Omega$

(C) $800\ \Omega$ and $0.32\ \Omega$ (D) $1.06\ \Omega$ and $500\ \Omega$

Q54. MCQ (Single Correct Answer)

Sun light falls normally on a surface of area $36\ \mathrm{cm}^2$ and exerts an average force of $7.2 \times 10^{-9}\ \mathrm{N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is

Open solution

(A) $25.92 \times 10^2\ \mathrm{W/cm}^2$ (B) $8.64 \times 10^{-6}\ \mathrm{W/cm}^2$

(C) $6.0\ \mathrm{W/cm}^2$ (D) $0.06\ \mathrm{W/cm}^2$

Q55. MCQ (Single Correct Answer)

The power of a lens (biconvex) is $1.25\ \mathrm{m}^{-1}$ in particular medium. Refractive index of the lens is 1.5 and radii of curvature are $20\ \mathrm{cm}$ and $40\ \mathrm{cm}$ respectively. The refractive index of surrounding medium:

Open solution

(A) 1.0 (B) $\frac{9}{7}$ (C) $\frac{3}{2}$ (D) $\frac{4}{3}$

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Q56. MCQ (Single Correct Answer)

Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be

(A) 1 : 2 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2

Open solution

Q57. Numerical

A ball is thrown vertically upwards with a velocity of $19.6\ \mathrm{ms}^{-1}$ from the top of a tower. The ball strikes the ground after $6\ \mathrm{s}$. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)\mathrm{m}$. The value of k is _____. (use $\mathrm{g} = 9.8\ \mathrm{m/s^2}$)

Open solution

Q58. Numerical

The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0\left(1 - \frac{x^2}{L^2}\right)\mathrm{kg/m}$ and length L (in meter) is $\frac{3L}{\alpha}\mathrm{m}$. The value of α is _____. (where x is the distance from end A)

Open solution

Q59. Numerical

A string of area of cross-section $4\ \mathrm{mm}^2$ and length $0.5\ \mathrm{m}$ is connected with a rigid body of mass $2\ \mathrm{kg}$. The body is rotated in a vertical circular path of radius $0.5\ \mathrm{m}$. The body acquires a speed of $5\ \mathrm{m/s}$ at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is _____ $\times 10^{-5}$.

(use Young's modulus $10^{11}\ \mathrm{N/m^2}$ and $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Open solution

Q60. Numerical

At a certain temperature, the degrees of freedom per molecule for gas is 8. The gas performs 150 J of work when it expands under constant pressure. The amount of heat absorbed by the gas will be _____ J.

Open solution

Q61. Numerical

The potential energy of a particle of mass $4\ \mathrm{kg}$ in motion along the x-axis is given by $\mathrm{U} = 4(1 - \cos 4x)$ J. The time period of the particle for small oscillation $(\sin\theta \simeq \theta)$ is $\left(\frac{\pi}{K}\right)s$. The value of K is _____.

Open solution

Q62. Numerical

An electrical bulb rated 220 V, 100 W, is connected in series with another bulb rated 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about _____ W.

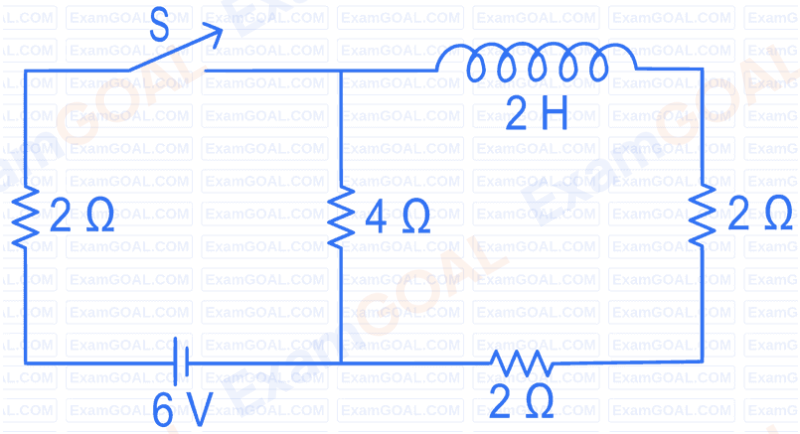
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Q63. Numerical

For the given circuit the current through battery of 6 V just after closing the switch 'S' will be _____ A.

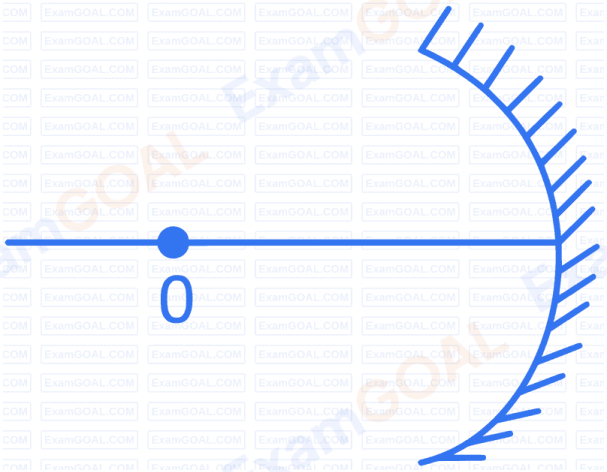
Open solution



Q64. Numerical

An object 'O' is placed at a distance of $100\ \text{cm}$ in front of a concave mirror of radius of curvature $200\ \text{cm}$ as shown in the figure. The object starts moving towards the mirror at a speed $2\ \text{cm/s}$. The position of the image from the mirror after $10\ \text{s}$ will be at _____ cm.

Open solution



Q65. Numerical

In an experiment with a convex lens, The plot of the image distance (v') against the object distance (μ') measured from the focus gives a curve $v'\mu' = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is _____ cm.

Open solution

Q66. Numerical

In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of $0.5\ \text{s}$ is measured from time of 100 oscillation with a watch of $1\ \text{s}$ resolution. If measured value of length is $10\ \text{cm}$ known to $1\ \text{mm}$ accuracy, The accuracy in the determination of g is found to be $x\%$. The value of x is _____.

Open solution

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Out of syllabus 24 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q67. MCQ (Single Correct Answer) · **Chemistry**

The correct decreasing order for metallic character is :

(A) Na > Mg > Be > Si > P

(B) P > Si > Be > Mg > Na

(C) Si > P > Be > Na > Mg

(D) Be > Na > Mg > Si > P

Open solution

Q68. MCQ (Single Correct Answer) · **Chemistry**

The products obtained during treatment of hard water using Clark's method are :

(A) $CaCO_3$ and $MgCO_3$

(B) $Ca(OH)_2$ and $Mg(OH)_2$

(C) $CaCO_3$ and $Mg(OH)_2$

(D) $Ca(OH)_2$ and $MgCO_3$

Open solution

Q69. MCQ (Single Correct Answer) · **Chemistry**

Statement I : An alloy of lithium and magnesium is used to make aircraft plates.

Statement II : The magnesium ions are important for cell-membrane integrity.

In the light the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Open solution

Q70. MCQ (Single Correct Answer) · **Chemistry**

White phosphorus reacts with thionyl chloride to give :

(A) PCl_5, SO_2 and S_2Cl_2

(B) PCl_3, SO_2 and S_2Cl_2

(C) PCl_3, SO_2 and Cl_2

(D) PCl_5, SO_2 and Cl_2

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

Concentrated HNO_3 reacts with Iodine to give :

(A) HI, NO_2 and H_2O

(B) HIO_2, N_2O and H_2O

(C) HIO_3, NO_2 and H_2O

(D) HIO_4, NO_2 and H_2O

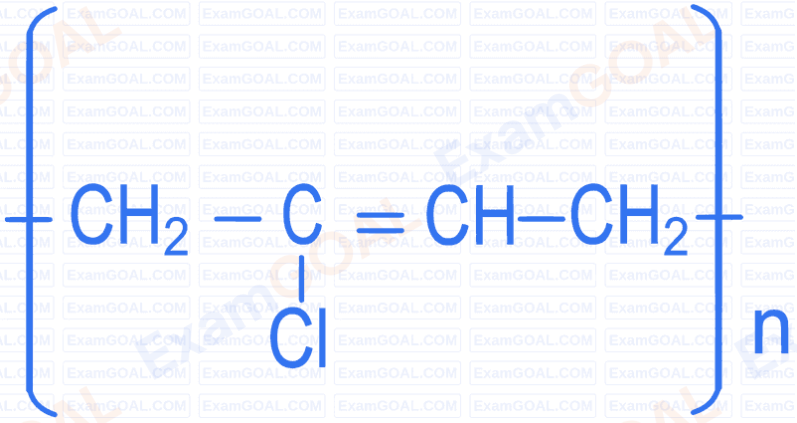
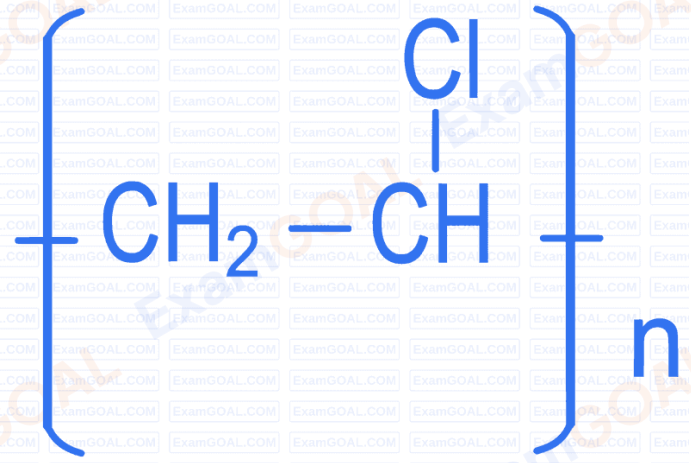
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Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List I with List II

	List - I (Polymer)		List - II (Nature)
(A)		(I)	Thermosetting polymer
(B)		(II)	Fibers
(C)		(III)	Elastomer
(D)		(IV)	Thermoplastic polymer

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Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I

(B) A-III, B-II, C-IV, D-I

(C) A-III, B-I, C-IV, D-II

(D) A-I, B-III, C-IV, D-II

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Two statements in respect of drug-enzyme interaction are given below

Statement I : Action of an enzyme can be blocked only when an inhibitor blocks the active site of the enzyme.

Statement II : An inhibitor can form a strong covalent bond with the enzyme.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q74. Numerical · **Chemistry**

Open solution

Metal M crystallizes into a fcc lattice with the edge length of $4.0 \times 10^{-8}\ \mathrm{cm}$. The atomic mass of the metal is _____ $\mathrm{g/mol}$. (Nearest integer)

(Use : $\mathrm{N_A} = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$, density of metal, $\mathrm{M} = 9.03\ \mathrm{g\ cm^{-3}}$)

Q75. Numerical · **Chemistry**

Open solution

On reaction with stronger oxidizing agent like $\mathrm{KIO_4}$, hydrogen peroxide oxidizes with the evolution of $\mathrm{O_2}$. The oxidation number of I in $\mathrm{KIO_4}$ changes to _____.

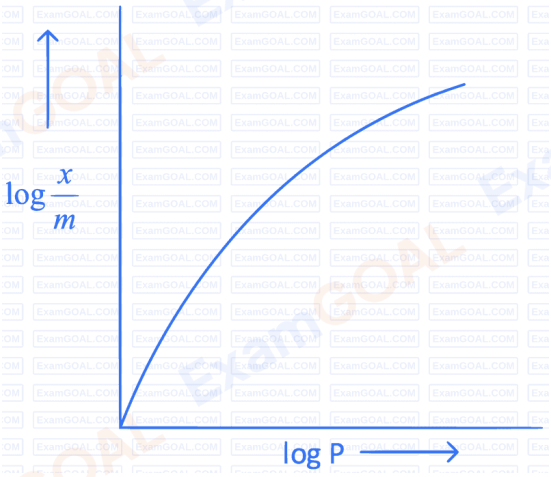
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Q76. Numerical · **Chemistry**

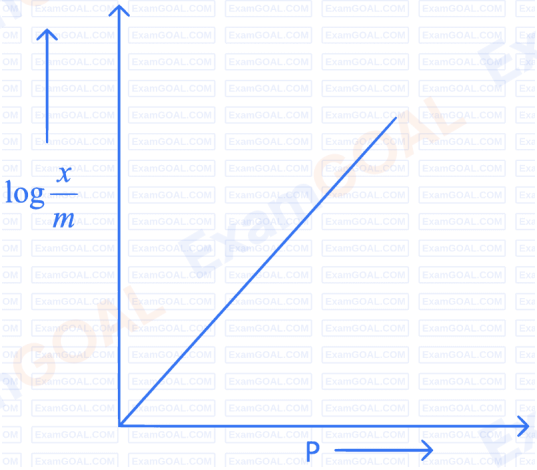
Open solution

Among the following the number of curves not in accordance with Freundlich adsorption isotherm is _____.

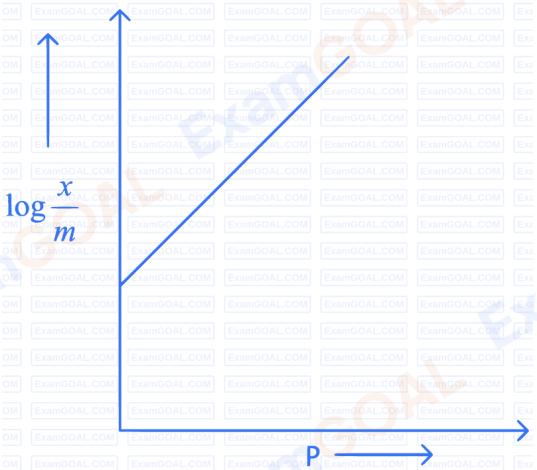
(a)



(b)

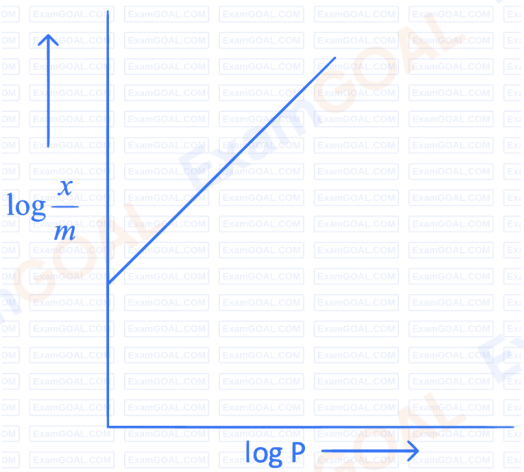


(c)



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(d)



Q77. MCQ (Single Correct Answer) · **Mathematics**

The differential equation of the family of circles passing through the points $(0, 2)$ and $(0, -2)$ is :

(A) $2xy\frac{dy}{dx} + \left(x^2 - y^2 + 4\right) = 0$

(B) $2xy\frac{dy}{dx} + \left(x^2 + y^2 - 4\right) = 0$

(C) $2xy\frac{dy}{dx} + \left(y^2 - x^2 + 4\right) = 0$

(D) $2xy\frac{dy}{dx} - \left(x^2 - y^2 + 4\right) = 0$

Open solution

Q78. MCQ (Single Correct Answer) · **Mathematics**

Let the tangents at two points A and B on the circle $x^2 + \mathrm{y}^2 - 4x + 3 = 0$ meet at origin $\mathrm{O}(0, 0)$. Then the area of the triangle OAB is :

(A) $\frac{3\sqrt{3}}{2}$

(B) $\frac{3\sqrt{3}}{4}$

(C) $\frac{3}{2\sqrt{3}}$

(D) $\frac{3}{4\sqrt{3}}$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let the lines

$\frac{x-1}{\lambda} = \frac{y-2}{1} = \frac{z-3}{2}$ and

$\frac{x+26}{-2} = \frac{y+18}{3} = \frac{z+28}{\lambda}$ be coplanar

and P be the plane containing these two lines.

Then which of the following points does **NOT** lie on P?

(A) $(0, -2, -2)$

(B) $(-5, 0, -1)$

(C) $(3, -1, 0)$

(D) $(0, 4, 5)$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

A plane P is parallel to two lines whose direction ratios are $-2, 1, -3$ and $-1, 2, -2$ and it contains the point $(2, 2, -2)$. Let P intersect the co-ordinate axes at the points $\mathrm{A}, \mathrm{B}, \mathrm{C}$ making the intercepts α, β, γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $\mathrm{p} = \alpha + \beta + \gamma$, then the ordered pair (V, p) is equal to :

(A) $(48, -13)$

(B) $(24, -13)$

(C) $(48, 11)$

(D) $(24, -5)$

Open solution

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Q81. MCQ (Single Correct Answer) · **Mathematics**

A horizontal park is in the shape of a triangle OAB with $\mathrm{AB} = 16$. A vertical lamp post OP is erected at the point O such that $\angle\mathrm{PAO} = \angle\mathrm{PBO} = 15^\circ$ and $\angle\mathrm{PCO} = 45^\circ$, where C is the midpoint of AB. Then $(\mathrm{OP})^2$ is equal to :

(A) $\frac{32}{\sqrt{3}}(\sqrt{3}-1)$ (B) $\frac{32}{\sqrt{3}}(2-\sqrt{3})$ (C) $\frac{16}{\sqrt{3}}(\sqrt{3}-1)$ (D) $\frac{16}{\sqrt{3}}(2-\sqrt{3})$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let

p : Ramesh listens to music.

q : Ramesh is out of his village.

r : It is Sunday.

s : It is Saturday.

Then the statement "Ramesh listens to music only if he is in his village and it is Sunday or Saturday" can be expressed as

(A) $((\sim q) \wedge (r \vee s)) \Rightarrow p$ (B) $(\mathrm{q} \wedge (\mathrm{r} \vee \mathrm{s})) \Rightarrow \mathrm{p}$

(C) $p \Rightarrow (q \wedge (r \vee s))$ (D) $\mathrm{p} \Rightarrow ((\sim \mathrm{q}) \wedge (\mathrm{r} \vee \mathrm{s}))$

Open solution

Q83. Numerical · **Mathematics**

Let the tangents at the points P and Q on the ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 1$ meet at the point $R(\sqrt{2}, 2\sqrt{2}-2)$. If S is the focus of the ellipse on its negative major axis, then $\mathrm{SP}^2 + \mathrm{SQ}^2$ is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

If $1 + (2 + {}^{49}C_1 + {}^{49}C_2 + \ldots + {}^{49}C_{49})({}^{50}C_2 + {}^{50}C_4 + \ldots + {}^{50}C_{50})$ is equal to $2^{\mathrm{n}} \cdot \mathrm{m}$, where m is odd, then $\mathrm{n} + \mathrm{m}$ is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

Two tangent lines l_1 and l_2 are drawn from the point $(2, 0)$ to the parabola $2\mathrm{y}^2 = -x$. If the lines l_1 and l_2 are also tangent to the circle $(x-5)^2 + y^2 = r$, then 17r is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \ldots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then m . n is equal to _____.

Open solution

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Q87. Numerical · **Mathematics**

Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$. Then the number of elements in the set $A = \{\theta \in S : \tan\theta(1 + \sqrt{5}\tan(2\theta)) = \sqrt{5} - \tan(2\theta)\}$ is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

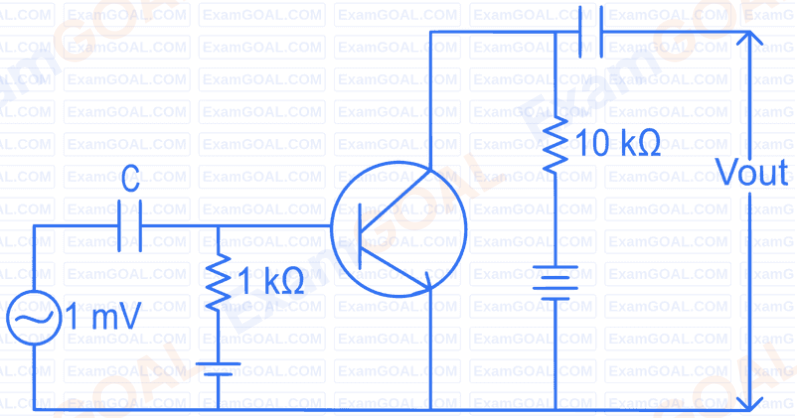
A radioactive sample decays $\frac{7}{8}$ times its original quantity in 15 minutes. The half-life of the sample is

(A) 5 min (B) 7.5 min (C) 15 min (D) 30 min

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

An n.p.n transistor with current gain $\beta = 100$ in common emitter configuration is shown in figure. The output voltage of the amplifier will be



(A) 0.1 V (B) 1.0 V (C) 10 V (D) 100 V

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A FM Broadcast transmitter, using modulating signal of frequency 20 kHz has a deviation ratio of 10. The Bandwidth required for transmission is :

(A) 220 kHz (B) 180 kHz (C) 360 kHz (D) 440 khz

Open solution

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Answer Key

Chemistry

1. **A**	2. **A**	3. **A, D**	4. **A**	5. **B**
6. **D**	7. **C**	8. **B**	9. **D**	10. **A**
11. **A**	12. **A**	13. **C**	14. **25**	15. **2**
16. **2**	17. **2**	18. **22**	19. **8**	20. **3**

Mathematics

21. **D**	22. **B**	23. **C**	24. **D**	25. **B**
26. **A**	27. **C**	28. **D**	29. **A**	30. **B**
31. **D**	32. **B**	33. **B**	34. **A**	35. **57**
36. **17**	37. **6**	38. **57**	39. **104**	

Physics

40. **D**	41. **B**	42. **D**	43. **D**	44. **A**
45. **C**	46. **C**	47. **C**	48. **A**	49. **B**
50. **C**	51. **C**	52. **C**	53. **C**	54. **D**
55. **B**	56. **C**	57. **392**	58. **8**	59. **30**
60. **750**	61. **2**	62. **14**	63. **1**	64. **400**
65. **15**	66. **5**			

Out of syllabus

67. **A**	68. **C**	69. **B**	70. **B**	71. **C**
72. **B**	73. **D**	74. **87**	75. **5**	76. **3**
77. **A**	78. **B**	79. **D**	80. **B**	81. **B**
82. **D**	83. **13**	84. **99**	85. **9**	86. **12**
87. **5**	88. **A**	89. **B**	90. **D**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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